

nginx JS — Dynamic Reconfiguration Guide (locations)

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DRAFT FOR COMMENTS — This document is a work in progress. Content, API names, and examples are subject to change before release.

1. Overview

This guide covers runtime management of nginx **location blocks** via the JavaScript DOM: adding, removing, cloning, and reconfiguring locations and servers without reloading nginx.

Prerequisites: read the *for single worker* guide first for `write_mode`, `snapshot()`, and `suspendAcceptance()`.

1.1. Scope

Location operations are available on:

- `NginxServer` objects — `nginx.http.servers[i]`
- `NginxLocation` objects — `nginx.http.servers[i].locations[j]` or the return value of `addLocation()`

They are **not** available on `req.location` (the request-phase wrapper is read-only for structural operations). Field writes (`root`, `handler`, ...) work on `req.location` subject to `write_mode`.

1.2. Config phase vs request phase

Location structure mutations (add, remove, clone) and handler assignment are permanent, global changes — they update the location BST immediately and affect every subsequent request on this worker. There is no per-request isolation for structural changes.

To propagate structural changes to all workers, use a `SharedWorker` broadcast combined with `suspendAllWorkers()` (see *for all workers* guide).

1.3. nginx location matching algorithm

Understanding the matching order is essential when adding locations:

```
Incoming URI
|
[]
1. Exact-match locations (= /path) ← checked first; wins immediately on match
|
[]
2. Prefix locations (longest wins among all prefix matches)
   |
   |— [] If longest match has ^~ ← skip regex scan; use this location
   |— [] Otherwise: remember the longest-prefix winner and continue ...
|
[]
3. Regex locations (checked in definition order, first match wins)
   |
   |— [] Match found ← use this location (overrides prefix winner)
   |— [] No match ← fall back to the remembered prefix winner
|
[]
4. Named locations (@name) – reachable only from error_page / try_files
```

This order determines which `addLocation()` patterns affect which requests.

2. Inspecting the location tree

2.1. `srv.locations[]`

Returns a live array of all locations registered on this server, including dynamically added ones. The order is: prefix/exact locations (sorted by path length, longest first), then regex locations (definition order), then named locations.

```
const srv = nginx.http.servers[0];

// List all locations with their match type and path
srv.locations.forEach(loc => {
  nginx.log(`${loc.matchType.padEnd(16)} ${loc.path}`);
});
```

`loc.matchType` values:

Value	Meaning
'prefix'	Normal longest-prefix match.
'exact'	<code>= /path</code> — checked first, wins immediately.
'prefix_noregex'	<code>^~ /path</code> — skips regex scan on match.
'regex'	<code>~ /pattern</code> — case-sensitive PCRE.
'regex_caseless'	<code>~* /pattern</code> — case-insensitive PCRE.
'named'	<code>@name</code> — reachable from <code>error_page</code> / <code>try_files</code> only.

`loc.path` — the full pattern string as passed to `addLocation()`, including the modifier prefix (e.g. `"=/exact"`, `"~ /\.\php$"`).

`loc.pattern` — the bare path/regex without the modifier prefix.

`loc.internal` — `true` for named and noname (`limit_except`, etc.) locations.

2.2. `srv.findLocation(pattern)`

Searches the server's location arrays by exact pattern string and returns the `NginxLocation` wrapper, or `null` if not found.

```
const srv = nginx.http.servers[0];

const api = srv.findLocation('/api/');
if (api) {
  nginx.log('api root: ' + api.root);
}
```

```
const exact = srv.findLocation('= /health');
const regex = srv.findLocation('~ /\.\.php$');
```

Uses the same pattern syntax as `addLocation()`.

2.3. Iterating and filtering

```
const srv = nginx.http.servers[0];

// All prefix locations with a JS handler
const jsLocs = srv.locations.filter(l =>
  l.matchType === 'prefix' && l.handler);

// All regex locations
const regexLocs = srv.locations.filter(l =>
  l.matchType === 'regex' || l.matchType === 'regex_caseless');

// Find by path prefix (e.g. all sub-locations of /api/)
const apiTree = srv.locations.filter(l =>
  l.matchType === 'prefix' && l.pattern.startsWith('/api/'));
```

3. Adding locations

3.1. `srv.addLocation(pattern [, opts])`

Adds a new location to the server and immediately rebuilds the location BST so the new location is active for all subsequent requests.

Returns the new `NginxLocation` object so properties and handler can be set in the same expression chain or immediately after.

```
const loc = srv.addLocation('/new/path');
loc.root    = '/var/www/new';
loc.handler = (req) => { req.sendFile('/index.html'); };
```

Pattern syntax:

Prefix	Example	Match type
(none)	<code>'/api/'</code>	Normal prefix — longest match wins.
<code>'= '</code>	<code>'= /ping'</code>	Exact match — wins immediately, highest priority after prefix search.
<code>'^~ '</code>	<code>'^~ /static/'</code>	Preferential prefix — disables regex scan when matched.
<code>'~ '</code>	<code>'~ /\.\.php\$'</code>	Case-sensitive PCRE regex.

Prefix	Example	Match type
'~*' '	'~* /\.\.jpg\$'	Case-insensitive PCRE regex.
'@'	'@fallback'	Named location — only reachable from <code>error_page</code> / <code>try_files</code> .

Options:

Option	Description
<code>template: '/path'</code>	Copy all config fields from an existing location (same server) as the starting point. The pattern of the new location is still <code>pattern</code> . Useful for creating near-copies of an existing location.
<code>index: N</code>	Regex locations only. Insert at position <code>N</code> in the regex check array (0 = first checked). Defaults to append (last checked).

Idempotency: if a location with the same pattern already exists, `addLocation()` returns the existing `NginxLocation` object without creating a duplicate. Safe to call multiple times.

3.2. `loc.addLocation(pattern [, opts])`

Delegates to the server-level `addLocation`. Available on locations obtained from `server.locations[]` or a prior `addLocation()` call. Not available on `req.location`.

```
// Equivalent – both add to the same server
srv.addLocation('/api/v2/');
apiLoc.addLocation('/api/v2/');    // apiLoc is any loc from this server
```



The BST builder automatically groups locations into a nested tree by shared prefix. You do not need to add child locations through a parent location object — adding `/api/v2/` to the server always nests it under `/api/` in the BST, regardless of which object you call `addLocation()` on.

3.3. Example: add a simple prefix location

```
// js_source: run at init_conf time

const srv = nginx.http.servers[0];
const docs = srv.addLocation('/docs/');

docs.root    = '/var/www/docs';
docs.sendfile = true;
docs.handler = (req) => {
  req.sendFile('/index.html');
};
```

3.4. Example: add an exact-match health endpoint

```
const health = srv.addLocation('= /health');
health.handler = (req) => {
  req.respond(200, { 'Content-Type': 'text/plain' }, 'OK');
};
```

Matching:

/health	—— exact match (= /health)	✓ highest priority
/health/more	—— prefix match /health	✓ (if /health exists)
/healthz	—— no match	□

3.5. Example: add a regex location at a specific position

```
const srv = nginx.http.servers[0];

// This regex will be checked BEFORE any existing regex locations
const php = srv.addLocation('~ /\.php$', { index: 0 });
php.handler = (req) => { /* dispatch to FastCGI */ };

// This regex will be checked LAST (default append)
const fallback = srv.addLocation('~* /\.html?$', { index: 1 });
fallback.root = '/var/www/html';
```

Regex check order (definition order):

[0]	~ /\.php\$	← index:0 → inserted first
[1]	~* /\.html?\$	← appended (default)
...	(existing regex locs)	

3.6. Example: add a named location for error fallback

```
const srv = nginx.http.servers[0];

const fallback = srv.addLocation('@static_fallback');
fallback.root = '/var/www/static';
fallback.handler = (req) => { req.sendFile(req.uri); };

// Wire it to 404 errors in another location
const api = srv.findLocation('/api/');
if (api) {
  api.errorPage = [{ codes: [404], uri: '@static_fallback' }];
}
```

```
}
```

3.7. Example: add a location using a template

```
const srv = nginx.http.servers[0];

// /api/v2/ gets the same config as /api/v1/ (handler, root, timeouts, ...)
// but at the new path
const v2 = srv.addLocation('/api/v2/', { template: '/api/v1/' });

// Override only what differs
v2.handler = v2Handler;
```

3.8. Flow diagram: addLocation BST rebuild

Before addLocation('/api/v2/')

Server location BST:

```
/          (prefix)
/api/      (prefix)
/api/v1/   (prefix, nested under /api/)
/static/   (prefix)
~ /\.php$  (regex[0])
```

addLocation('/api/v2/') called:

1. New clcf allocated in dyn_pool
2. entry pushed to prefix_locs[]
3. ngx_js_rebuild_loc_tree() called → new BST built from ALL prefix_locs[]
4. srv->root->static_locations = new BST root
5. New requests immediately use the new BST

After:

```
/          (prefix)
/api/      (prefix)
/api/v1/   (prefix, nested under /api/)
/api/v2/   (prefix, nested under /api/) ← NEW
/static/   (prefix)
~ /\.php$  (regex[0])
```

4. Removing locations

4.1. srv.removeLocation(pattern)

Removes the location matching **pattern** from the server. The pattern syntax is the same as

`addLocation()`. Returns `true` if found and removed, `false` if not found.

The BST is rebuilt immediately after removal so the location stops matching new requests right away.

```
const removed = srv.removeLocation('/api/v1/');
nginx.log('removed: ' + removed); // true or false
```

```
// Remove a regex location
const ok = srv.removeLocation('~ /\.\.php$');

// Remove an exact-match location
srv.removeLocation('= /health');

// Remove a named location
srv.removeLocation('@fallback');
```



`removeLocation` matches by pattern string only — the same string you passed to `addLocation`. It does not perform URI matching.

4.2. Example: maintenance mode toggle

Add a catch-all location for maintenance and remove it when done.

```
const srv = nginx.http.servers[0];
let maintenanceActive = false;

function enableMaintenance() {
  if (maintenanceActive) { return; }

  nginx.suspendAcceptance();

  const m = srv.addLocation('/'); // catches everything (longest prefix: /)
  m.handler = (req) => {
    req.respond(503, { 'Retry-After': '120' },
      'Service temporarily unavailable');
  };

  nginx.resumeAcceptance();
  maintenanceActive = true;
}

function disableMaintenance() {
  if (!maintenanceActive) { return; }

  nginx.suspendAcceptance();
  srv.removeLocation('/');
```

```

    nginx.resumeAcceptance();

    maintenanceActive = false;
}

```

4.3. Example: remove all dynamically added locations

```

// Remove every location whose path starts with /dynamic/
srv.locations
    .filter(l => l.pattern.startsWith('/dynamic/'))
    .forEach(l => srv.removeLocation(l.path));

```



Removing a location while iterating `srv.locations` is safe because `srv.locations` returns a snapshot array; it is not mutated during iteration.

4.4. Flow diagram: removeLocation BST rebuild

Before `removeLocation('/api/v1/')`:

```

prefix_locs[]:  /, /api/, /api/v1/, /api/v2/, /static/
BST:
    /
   ├── /api/
   │   ├── /api/v1/    ← to be removed
   │   └── /api/v2/
   └── /static/

```

`removeLocation('/api/v1/')`:

1. entry spliced out of `prefix_locs[]`
2. `ngx_js_rebuild_loc_tree()` → new BST
3. New requests immediately stop matching `/api/v1/`
4. In-flight requests already dispatched to `/api/v1/` complete normally

After:

```

BST:
    /
   ├── /api/
   │   └── /api/v2/
   └── /static/

```



In-flight requests that were already dispatched to a location before it is removed will complete normally. The removed location's handler and config remain valid for the duration of those requests.

5. Cloning locations

5.1. `loc.clone(newPattern [, opts])`

Creates a new prefix location at `newPattern` as a shallow copy of the source location — same handler, same config values (root, timeouts, proxy settings, etc.). Only plain prefix patterns (no `=`, `^~`, `~`, `@`) are supported as `newPattern`.

Option	Description
<code>depth: N</code>	Copy child locations up to <code>N</code> levels of relative nesting. <code>0</code> (default when omitted) = only the location itself. <code>Infinity</code> = copy all descendants.

Returns the new `NginxLocation` object. Idempotent: if `newPattern` already exists as a prefix location, returns the existing wrapper.

```
const src = srv.findLocation('/api/v1/');
const dst = src.clone('/api/v2/');

// dst is an independent location – mutating it does not affect src
dst.handler = v2Handler;
```

5.2. Example: versioned API clone

```
const srv = nginx.http.servers[0];
const v1 = srv.findLocation('/api/v1/');

// Clone /api/v1/ tree to /api/v2/ (all immediate children)
const v2 = v1.clone('/api/v2/', { depth: 1 });

// Before:  /api/v1/, /api/v1/users, /api/v1/orders
// After:   all of the above PLUS:
//          /api/v2/, /api/v2/users, /api/v2/orders

// Override the v2 root handler
v2.handler = (req) => {
  req.respond(200, {}, 'API v2');
};
```

Source tree (/api/v1/):

```
/api/v1/
/api/v1/users
/api/v1/orders
```

After `clone('/api/v2/', { depth: 1 })`:

```
/api/v1/          (unchanged)
```

/api/v1/users	(unchanged)
/api/v1/orders	(unchanged)
/api/v2/	(new, copy of /api/v1/)
/api/v2/users	(new, copy of /api/v1/users)
/api/v2/orders	(new, copy of /api/v1/orders)

5.3. Example: full subtree clone with depth:Infinity

```
const v1 = srv.findLocation('/api/v1/');

// Copies all descendants regardless of nesting depth
const v3 = v1.clone('/api/v3/', { depth: Infinity });

// Customise only the top level
v3.root = '/var/www/v3';
```

5.4. Example: clone a single location, no children

```
const beta = srv.findLocation('/beta/');

// depth defaults to 0: only /beta/ itself is cloned, not /beta/api/ etc.
const prod = beta.clone('/prod/');
prod.root = '/var/www/prod';
```

6. Cloning a server

6.1. srv.clone(newServerName)

Creates a shallow copy of the entire server — all locations, all config fields — and registers it as a new virtual host under `newServerName`. The clone is added to `nginx.http.servers[]` and starts accepting requests for `newServerName` immediately.

```
const primary = nginx.http.servers.find(s => s.name === 'example.com');
const mirror = primary.clone('mirror.example.com');

// Customise the clone
mirror.root = '/var/www/mirror';
```



`srv.clone()` creates independent location arrays and an independent BST for the new server. Structural changes on the clone do not affect the original, and vice versa.

6.2. Example: per-tenant server provisioning

```
const template = nginx.http.servers.find(s => s.name === 'template.internal');

const tenants = ['acme', 'globex', 'initech'];
tenants.forEach(tenant => {
  const srv = template.clone(tenant + '.example.com');
  srv.root = '/var/www/tenants/' + tenant;

  // Add tenant-specific admin location
  const admin = srv.addLocation('= /admin');
  admin.handler = makeAdminHandler(tenant);
});
```

7. Modifying existing locations

7.1. Changing handler and properties

Property writes on a location obtained from `server.locations[]` or `addLocation()` use `write_mode='global'` by default — they affect all future requests on this worker immediately.

```
const loc = srv.findLocation('/api/');
if (loc) {
  loc.root    = '/var/www/api-v2';
  loc.handler = newApiHandler;
}
```

7.2. Atomic batch modification

Use `suspendAcceptance()` when changing multiple properties to prevent any new request on this worker from seeing a half-updated location:

```
const loc = srv.findLocation('/api/');

nginx.suspendAcceptance();

loc.root          = '/var/www/new';
loc.keepaliveTimeout = 75000;
loc.proxy.readTimeout  = 30000;
loc.proxy.connectTimeout = 5000;

nginx.resumeAcceptance();
```

7.3. Rollback with snapshot

```
const loc = srv.findLocation('/api/');
const snap = loc.snapshot();

nginx.suspendAcceptance();
try {
  loc.root = cfg.root;
  loc.handler = cfg.handler;
  nginx.resumeAcceptance();
  req.json({ applied: true });
} catch (e) {
  snap.restore();
  nginx.resumeAcceptance();
  req.respond(500, {}, e.message);
}
```

7.4. Location swap (atomic replace)

Replace one location with another in a single suspension window:

```
async function swapLocations(srv, oldPattern, newPattern, newHandler, newRoot) {
  nginx.suspendAcceptance();

  try {
    srv.removeLocation(oldPattern);

    const loc = srv.addLocation(newPattern);
    loc.handler = newHandler;
    loc.root = newRoot;
  } finally {
    nginx.resumeAcceptance();
  }
}

// Example: swap /beta/ → /prod/
await swapLocations(srv, '/beta/', '/prod/', prodHandler, '/var/www/prod');
```

Before swap:		After swap:	
/beta/	(active)	/prod/	(active, immediate)
/prod/	(absent)	/beta/	(absent)

8. Atomic location changes across all workers

8.1. The problem

`addLocation()`, `removeLocation()`, `clone()`, and property writes all update only the **current worker's** location BST. Other workers continue to use their own BST, which does not include the change.

To apply structural changes consistently across all workers, combine `suspendAllWorkers()` with a `SharedWorker` broadcast.

8.2. Pattern: broadcast structural change

```
// js_source: location_mgr.js

const cfgWorker = new SharedWorker('cfgworker.js');

cfgWorker.onmessage = (msg) => {
  const { op, pattern, opts, props } = msg.data;
  const srv = nginx.http.servers[0];

  if (op === 'add') {
    const loc = srv.addLocation(pattern, opts || {});
    if (props) {
      if (props.root) { loc.root = props.root; }
      if (props.handler) { loc.handler = eval(props.handler); } // see note
    }
  } else if (op === 'remove') {
    srv.removeLocation(pattern);
  }
};
```

```
// Request handler: add a location on all workers

loc.handler = async (req) => {
  const cfg = JSON.parse(await req.readBody());

  // Apply on this worker + broadcast to all others
  await nginx.suspendAllWorkers();

  try {
    const newLoc = srv.addLocation(cfg.pattern);
    newLoc.root = cfg.root;
    newLoc.handler = makeHandler(cfg);

    // Broadcast (other workers apply in their onmessage)
```

```

    cfgWorker.postMessage({ op: 'add', pattern: cfg.pattern,
                           props: { root: cfg.root } });

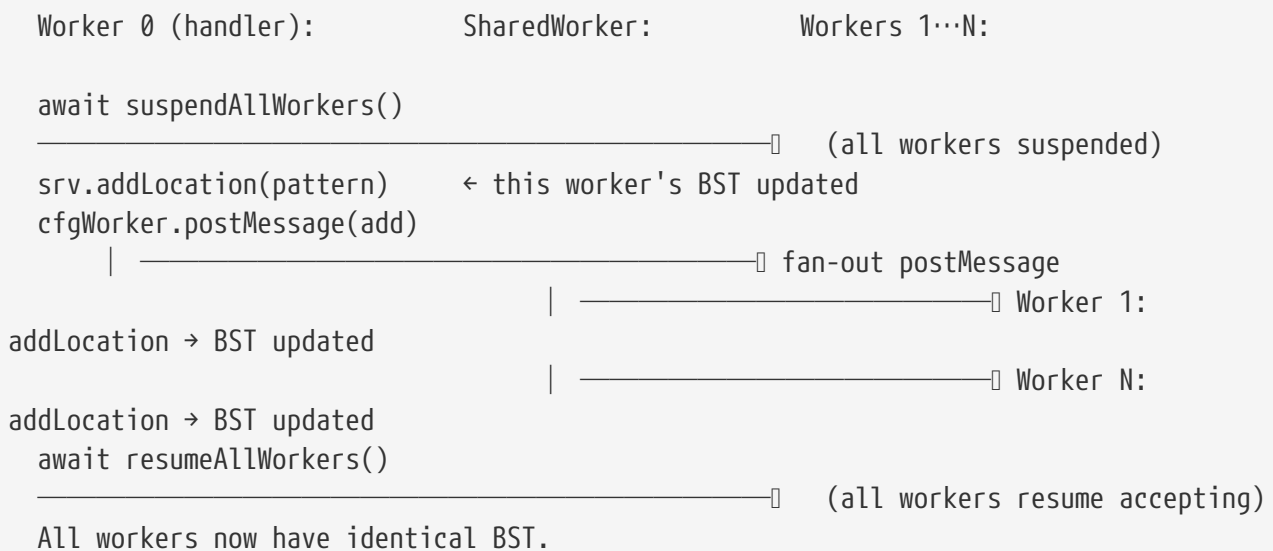
    req.json({ added: cfg.pattern });
  } finally {
    await nginx.resumeAllWorkers();
  }
};

```



JS handler functions cannot be serialised via `postMessage`. For broadcast, pass a handler identifier string and let each worker resolve it to a locally defined function (e.g. via a registry map).

8.3. Flow diagram: cross-worker addLocation



9. Safe patterns

9.1. Idempotent add-or-update

```

function addOrUpdate(srv, pattern, root, handler) {
  // addLocation is idempotent – returns existing loc if pattern exists
  const loc = srv.addLocation(pattern);
  loc.root = root;
  loc.handler = handler;
  return loc;
}

```

9.2. Temporary location (feature flag)

Add a location when a feature is enabled, remove it when disabled.

```
const FLAG_PATH = '/api/new-feature/';
let featureEnabled = false;

function setFeature(enabled) {
  if (enabled === featureEnabled) { return; }

  nginx.suspendAcceptance();

  if (enabled) {
    const loc = srv.addLocation(FLAG_PATH);
    loc.handler = newFeatureHandler;
  } else {
    srv.removeLocation(FLAG_PATH);
  }

  nginx.resumeAcceptance();
  featureEnabled = enabled;
}
```

9.3. Versioned API routing

```
// js_source: versions.js

const srv = nginx.http.servers[0];

// V1 is defined statically in nginx.conf
const v1 = srv.findLocation('/api/v1/');

// V2 is a clone of V1, handler overridden
const v2 = v1.clone('/api/v2/', { depth: Infinity });
v2.handler = v2ApiHandler;

// V3 is freshly added with no template
const v3 = srv.addLocation('/api/v3/');
v3.root = '/var/www/api/v3';
v3.handler = v3ApiHandler;
```

9.4. Location inventory endpoint

Expose the current location tree as JSON for observability:

```
const inv = srv.addLocation('= /__inventory');
```

```

inv.handler = (req) => {
  const locs = srv.locations.map(l => ({
    path:      l.path,
    matchType: l.matchType,
    root:      l.root,
    hasHandler: !!l.handler,
    internal:  l.internal,
  }));
  req.json({ locations: locs, count: locs.length });
};

```

9.5. A/B routing by location clone

Route a percentage of traffic to a cloned experimental location:

```

// js_source: ab.js

const srv    = nginx.http.servers[0];
const stable = srv.findLocation('/api/');
const exp    = stable.clone('/api/__exp/');
exp.handler  = experimentalApiHandler;

// Router: override location choice per-request
const router = srv.addLocation('/api/');
const origHandler = stable.handler;

router.handler = (req) => {
  if (Math.random() < 0.05) {
    // Internally reroute 5% to experimental handler
    experimentalApiHandler(req);
  } else {
    origHandler(req);
  }
};

```

10. API Reference

10.1. `srv.locations`

Read-only array of all `NginxLocation` objects registered on this server. Includes static (from `nginx.conf`) and dynamic (added by `addLocation()`) locations. Order: prefix/exact (longest path first), then regex (definition order), then named.

```
NginxLocation[]  srv.locations
```

10.2. `srv.findLocation(pattern)`

Returns the `NginxLocation` for the given `pattern` string, or `null` if not found. Searches prefix/exact, regex, and named arrays.

```
NginxLocation | null  srv.findLocation(pattern)
```

Pattern syntax is identical to `addLocation()`.

10.3. `srv.addLocation(pattern [, opts])`

Adds a new location (or returns existing if `pattern` already registered). Rebuilds the location BST immediately.

```
NginxLocation  srv.addLocation(pattern)
NginxLocation  srv.addLocation(pattern, { template: '/path', index: N })
```

Throws	Condition
<code>TypeError</code>	<code>pattern</code> is not a string; regex pattern on a build without PCRE.
<code>InternalError</code>	BST rebuild failed (OOM).

10.4. `srv.removeLocation(pattern)`

Removes the location matching `pattern` and rebuilds the BST.

```
boolean  srv.removeLocation(pattern)
```

Returns `true` if found and removed, `false` if not found.

10.5. `srv.clone(newServerName)`

Creates a full shallow copy of the server under `newServerName`, adds it to `nginx.http.servers[]`.

```
NginxServer  srv.clone(newServerName)
```

10.6. `loc.addLocation(pattern [, opts])`

Identical to `srv.addLocation()` — delegates to the server. Not available on `req.location`.

```
NginxLocation  loc.addLocation(pattern)
```

```
NginxLocation loc.addLocation(pattern, { template: '/path', index: N })
```

10.7. loc.removeLocation(pattern)

Identical to `srv.removeLocation()` — delegates to the server. Not available on `req.location`.

```
boolean loc.removeLocation(pattern)
```

10.8. loc.clone(newPattern [, opts])

Creates a shallow copy of this location at `newPattern` (plain prefix only). Optionally copies child locations up to `depth` levels deep.

```
NginxLocation loc.clone(newPattern)
NginxLocation loc.clone(newPattern, { depth: N })
```

depth	Effect
0 (default)	Only this location; no children.
N	Children up to N levels of relative nesting depth.
Infinity	All descendants.

Throws `TypeError` if `newPattern` has a modifier prefix (`=`, `^~`, `~`, `@`).

10.9. loc.path, loc.matchType, loc.pattern, loc.internal

Read-only properties on every `NginxLocation`:

Property	Type	Description
path	string	Full pattern including modifier (e.g. <code>"= /ping"</code> , <code>"~ /*.php\$"</code>).
matchType	string	One of: <code>'prefix'</code> , <code>'exact'</code> , <code>'prefix_noregex'</code> , <code>'regex'</code> , <code>'regex_caseless'</code> , <code>'named'</code> .
pattern	string	Bare path/regex without modifier prefix.
internal	boolean	<code>true</code> for named and internal (noname) locations.

10.10. nginx.suspendAcceptance() / nginx.resumeAcceptance()

Synchronous per-worker acceptance control. Use to wrap multi-step location mutations that must appear atomic to new requests on this worker. See the *for single worker* guide for full

documentation.

10.11. `nginx.suspendAllWorkers()` / `nginx.resumeAllWorkers()`

Async cross-worker acceptance control. Returns a `Promise<undefined>`. Use when location changes must be visible consistently across all workers before any new request is accepted. See the *for all workers* guide for full documentation.

11. Design notes

11.1. BST rebuild cost

Every `addLocation()`, `removeLocation()`, or `clone()` call rebuilds the location binary search tree from the flat `prefix_locs[]` snapshot. The rebuild is $O(N \log N)$ in the number of prefix locations. It allocates a fresh pool for the BST nodes and releases the previous one.

For typical deployments (tens to hundreds of locations) the rebuild is measured in microseconds and is not a concern. For automated bulk operations (thousands of locations), batch them in a single suspension window to amortise the rebuild cost.

11.2. Dynamic pool vs static pool

Config-file locations are allocated in nginx's main memory pool (`cycle→pool`), which is never freed during a reload. Dynamically added locations are allocated in the server's `dyn_pool`, which is freed when the server wrapper is garbage-collected (typically at process exit or explicit release).

Removed locations are spliced out of the live arrays but their memory in `dyn_pool` is not reclaimed until the pool is destroyed. This is harmless for typical usage patterns.

11.3. `r.location` is structural read-only

`req.location` provides a per-request view of the matched location. `addLocation()`, `removeLocation()`, and `clone()` are not available on it. Field writes (`root`, `handler`, etc.) work subject to `write_mode`.

The rationale: structural changes are global operations on the server's BST; they would not make sense in a per-request isolation context.

11.4. Regex position and definition order

The `index` option in `addLocation('~ pattern', { index: N })` controls where the new regex is inserted in the regex array. Regex locations are always checked in array order; a lower index means higher priority.

Removing a regex location compacts the array: all locations after the removed entry shift down by one position.

11.5. Handler serialisation for cross-worker broadcast

JS functions cannot be transferred via `postMessage`. For cross-worker location provisioning, use a handler registry: map handler names to function references, broadcast the name, and let each worker look it up locally.

```
// Shared handler registry (defined in js_source, visible in every worker)
const handlerRegistry = {
  'api-v2': v2ApiHandler,
  'api-v3': v3ApiHandler,
  'static': staticHandler,
  'maintenance': maintenanceHandler,
};

// On receiving a broadcast:
cfgWorker.onmessage = (msg) => {
  const loc = srv.addLocation(msg.data.pattern);
  const handler = handlerRegistry[msg.data.handlerName];
  if (handler) { loc.handler = handler; }
};
```